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Concept Review

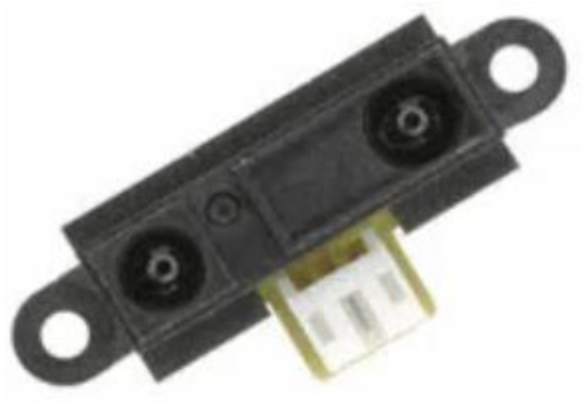
Infrared Distance

Why Do We Need Infrared Distance Sensors?

When designing an electromechanical system, we consider electrical requirements, mechanical complexity and environmental impact on the sensor's measurement integrity. Infrared distance sensors emit infrared pulses and measure the reflected infrared energy. Mechanically, they depend on a lens to focus light for both the emitter and detector. Infrared distance sensors can be found in robotics, aviation, smartphones, etc..

Infrared Distance

Using the same concept as Time-of-Flight sensors, infrared distance sensors use emitters and receivers to convert infrared radiation into an analog distance measurement. Figure 1. shows examples of different models/types of Infrared distance sensors.



Sharp GP2Y0A21YK0F

Infrared Pair Tube

Figure 1: Different configurations for infrared distance sensors.

Figure 2 shows an example of a laser-based time of flight sensor which estimates the straight-line distance to a subject of interest.

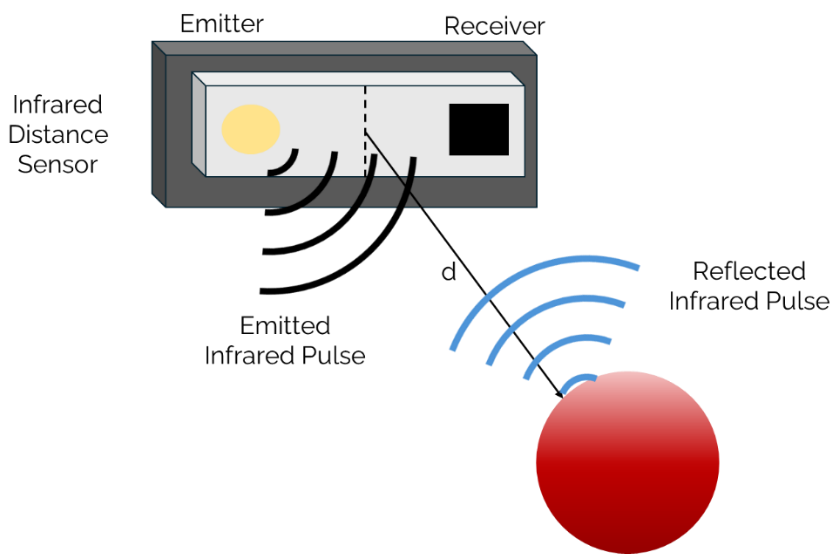


Figure 2: Distance estimate from an Infrared Distance sensor

A mechanical difference between sound focused distance sensors (such as ultrasonic) which use elements such as acoustic cones to focus sound, infrared distance sensors use lenses to focus the reflected infrared wave.

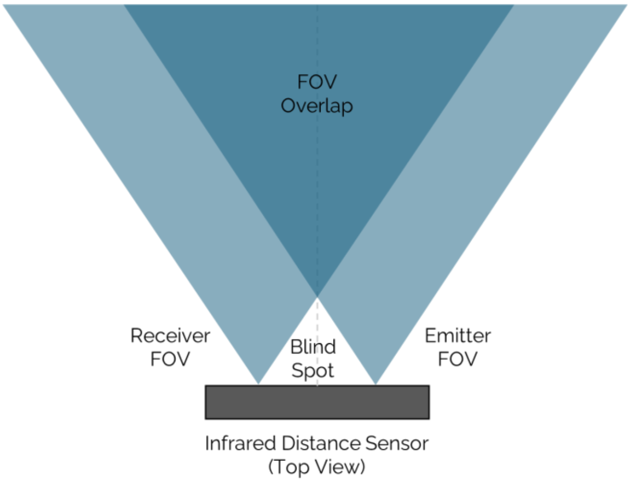


Figure 2: Lens field of view overlap.

With the use of lenses to focus pulsated infrared waves, distance measurements have a working range. As an example, for the Sharp GP2Y0A21YK0F the valid distance measurement range is from 10cm up to 80 cm. Another important characteristic of infrared distance sensors is the effect of infrared energy dissipation as a function of distance. As an example, Figure 3. shows the relationship between distance to a reflective object vs Output Voltage for the Sharp GP2Y0A21YK0F.

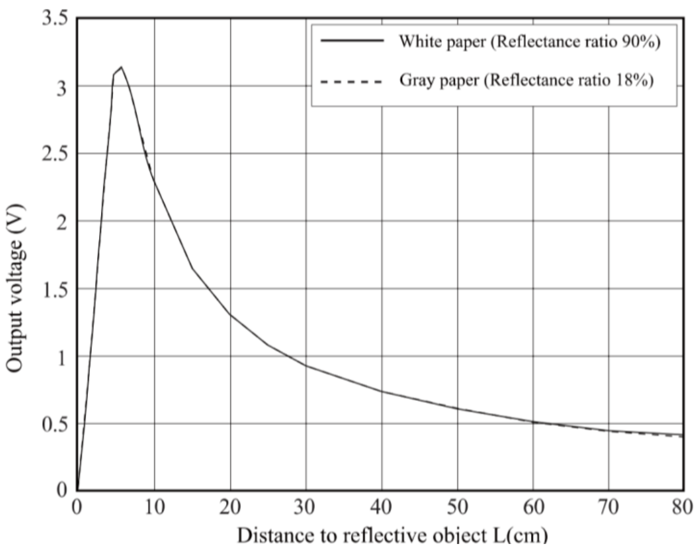


Figure 3. Distance to reflective object vs Output voltage

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